

38. Jointly with R. Brauer and H. Hasse: Proof of a Main Theorem in the Theory of Algebras

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At last our joint efforts have succeeded in proving the following theorem, which is of fundamental importance for the structure theory of algebras over algebraic number fields and beyond.

Main Theorem. Every normal division algebra over an algebraic number field is cyclic, or, equivalently, of Dickson type.

It is a special pleasure to present this result, essentially an achievement of the p -adic method, to Kurt Hensel, the founder of that method, on his seventieth birthday. Our proof consists of three reductions, each contributed by one of us.

H. Hasse gave the first reduction on the basis of his recently developed theory of cyclic algebras over algebraic number fields.

Reduction 1. The main theorem is proved once the following is shown:

I. Every everywhere split algebra over Ω is $\sim \Omega$.

Here and below, an “algebra A over Ω ” means a normal simple algebra over the algebraic number field Ω , i.e. a simple hypercomplex system with center Ω . The notation $A \sim \Omega$ means that A is a full matrix algebra over Ω , equivalently that it belongs to the special similarity class determined by Ω itself as division algebra. An algebra is called everywhere split if for every prime place $\mathfrak{p}$ of Ω its $\mathfrak{p}$ -adic extension $A_{\mathfrak{p}}$ is split:

$$A_{\mathfrak{p}} \sim \Omega_{\mathfrak{p}}.$$

Let D be a division algebra over Ω . Choose a cyclic field Z/Ω such that, for every prime place $\mathfrak{p}$, the $\mathfrak{p}$ -degree of Z is a multiple of the $\mathfrak{p}$ -index of D . Then for every prime divisor $\mathfrak{P}$ of $\mathfrak{p}$ in Z , the local field $Z_{\mathfrak{P}}$ splits $D_{\mathfrak{p}}$. Hence the algebra D_Z obtained by extending scalars from Ω to Z is everywhere split. If I is known, $D_Z \sim Z$. Thus D has a cyclic splitting field Z , hence is cyclically representable; moreover it has such a cyclic splitting field whose degree agrees with the degree of D . Therefore D is cyclic.

The second reduction was prompted decisively by R. Brauer, who informed H. Hasse by letter that the related question of the exact value of the exponent of a division algebra can be reduced, using Sylow’s theorem, to the case of a solvable splitting field. The same idea gives the next reduction of the cyclicity question.

Reduction 2. Statement I is proved once the following is shown:

II. Every everywhere split algebra over Ω with a solvable splitting field is $\sim \Omega$.

Let A be everywhere split over Ω . Let K be a Galois splitting field for A , let p be a prime number, and let Σ be the Sylow field of K/Ω corresponding to p , i.e. the fixed field of a p -Sylow subgroup of the Galois group. Then A_{Σ} is an everywhere split algebra over Σ , and has the solvable splitting field K . By II, $A_{\Sigma} \sim \Sigma$. Thus Σ is a splitting field of A , and its degree over Ω is prime to p . Hence the index of A , being a divisor of that degree, is prime to p . Since this holds for every prime p , the index is 1, and $A \sim \Omega$.

The third reduction, which together with the first two leads to the final proof, was extracted by E. Noether from a communication of Hasse; independently, Brauer had also considered it.

Reduction 3. Statement II is proved once the following is shown:

III. Every everywhere split algebra over Ω with a cyclic splitting field of prime degree is $\sim \Omega$.

Let A be everywhere split and have a solvable splitting field K/Ω . Choose a tower

$$K = \Lambda_0 > \Lambda_1 > \cdots > \Lambda_r = \Omega$$

such that each Λ_i/Λ_{i+1} is cyclic of prime degree. Then A_{Λ_1} is everywhere split over Λ_1 , and has the cyclic splitting field $K = \Lambda_0$ of prime degree. By III, $A_{\Lambda_1} \sim \Lambda_1$, so Λ_1 already splits A . Repeating the same argument down the tower gives $A \sim \Omega$. Thus II follows from III.

But III is known from Hasse's results. Hence the main theorem follows by reversing the reductions 3, 2, and 1. Noether further observes that III, in the more general form for cyclic splitting fields of arbitrary degree, is equivalent to Hasse's norm theorem. For Reduction 3 one needs only the prime-degree case, the original Hilbert–Furtwängler norm theorem. Thus Reduction 3 gives a new simple proof of Hasse's norm theorem. In Hasse's notation: if Z/Ω is cyclic and $\alpha \in \Omega$ satisfies

$$\left(\frac{\alpha, Z}{\mathfrak{p}}\right) = 1$$

for every prime place $\mathfrak{p}$, then every cyclic algebra

$$A = (\alpha, Z)$$

is everywhere split. Reduction 3 gives $A \sim \Omega$, and hence α is the norm of an element of Z .

Consequences

Theorem 1. The exponent of a normal simple algebra over an algebraic number field is equal to its index.

Theorem 2. The fundamental ideal of a proper normal division algebra over Ω is divisible by at least one prime place of Ω , and in fact by at least two.

The fundamental ideal may be defined as the reduced norm of the reduced different. An infinite prime place is included exactly when the algebra reduces there to the quaternion algebra, rather than to the real or complex number field. By Hasse's results, a prime place occurs in the fundamental ideal exactly when the $\mathfrak{p}$ -index is not 1. If all $\mathfrak{p}$ -indices were 1, the algebra would be everywhere split; then by I it would not be a proper division algebra. Nor can a single $\mathfrak{p}$ -index alone be different from 1, because the norm residue symbols, whose orders are the local indices, satisfy the product theorem.

Theorem 3. For an algebra A over Ω , an algebraic number field K/Ω is a splitting field if and only if, for every prime divisor $\mathfrak{P}_i$ in K of every prime place $\mathfrak{p}$ of Ω , the $\mathfrak{P}_i$ -degree $n_{\mathfrak{P}_i}$ of K is a multiple of the $\mathfrak{p}$ -index $m_{\mathfrak{p}}$ of A .

If

$$\mathfrak{p} = \mathfrak{P}_1^{e_{\mathfrak{P}_1}} \cdots \mathfrak{P}_g^{e_{\mathfrak{P}_g}}, \quad N(\mathfrak{P}_i) = \mathfrak{p}^{f_{\mathfrak{P}_i}},$$

then

$$n_{\mathfrak{P}_i} = e_{\mathfrak{P}_i} f_{\mathfrak{P}_i}.$$

The $\mathfrak{p}$ -index of A is the index $m_{\mathfrak{p}}$ of $A_{\mathfrak{p}}$.

The assertion follows because K splits A precisely when $A_{K_{\mathfrak{P}_i}} \sim K_{\mathfrak{P}_i}$ for all $\mathfrak{P}_i$. At a finite prime place let

$$A_{\mathfrak{p}} \sim (\alpha, W_{\mathfrak{p}})$$

be the distinguished cyclic local representation, where $W_{\mathfrak{p}}$ is the unramified field of degree $m_{\mathfrak{p}}$ over $\Omega_{\mathfrak{p}}$. Let B_i be the intersection and C_i the composite of $W_{\mathfrak{p}}$ and $K_{\mathfrak{P}_i}$. With

$$d_i = (m_{\mathfrak{p}}, f_{\mathfrak{P}_i}),$$

the extension $B_i/\Omega_{\mathfrak{p}}$ has degree d_i , and $K_{\mathfrak{P}_i}/B_i$ is unramified of degree $m_{\mathfrak{p}}/d_i$. The local algebra splits exactly when α is a norm from $K_{\mathfrak{P}_i}$ over B_i , which is equivalent to the valuation of α in $\mathfrak{P}_i$ being a multiple of $m_{\mathfrak{p}}/d_i$. Since

$$\left(\frac{m_{\mathfrak{p}}}{d_i}, \frac{f_{\mathfrak{P}_i}}{d_i}\right) = 1,$$

this is equivalent to $n_{\mathfrak{P}_i}$ being a multiple of $m_{\mathfrak{p}}$. For infinite prime places, unless $m_{\mathfrak{p}} = 1$, one has $m_{\mathfrak{p}} = 2$, and the condition says precisely that the corresponding local field is complex.

Theorem 4. *Decomposition theorem.* Let K/Ω be Galois, and let $\mathfrak{p}$ be a prime ideal of Ω not dividing the relative discriminant of K . The relative degree f of the prime divisors of $\mathfrak{p}$ in K is the least exponent for which

$$A^f \sim \Omega$$

for every algebra A in the subgroup of Brauer's group assigned to K .

Indeed, f is the local degree of K at $\mathfrak{p}$, while the $\mathfrak{p}$ -index of A is the index, equivalently the exponent, of $A_{\mathfrak{p}}$; Theorem 3 supplies the local criterion. The existence of algebras with exact $\mathfrak{p}$ -index f follows from Frobenius' density theorem and the construction of a cyclic algebra (α, Z) whose norm residue symbols at two suitably chosen prime places have reciprocal values of order f and are trivial elsewhere.

Theorem 5. *Uniqueness and ordering theorem.* $K \subseteq K'$ if and only if $\mathfrak{R}_K \subseteq \mathfrak{R}_{K'}$. In particular, the correspondence between Galois fields K and their algebra groups is one-to-one and order-preserving.

The implication $K \subseteq K' \Rightarrow \mathfrak{R}_K \subseteq \mathfrak{R}_{K'}$ is immediate. Conversely, the decomposition theorem gives the required divisibilities of relative degrees for all primes outside the relative discriminant; the standard analytic argument then yields $K \subseteq K'$.

Theorem 6. All absolutely irreducible representations of a finite group G can be realized over cyclotomic fields; in any case they can be realized over the field of n^h -th roots of unity for $n = |G|$ and sufficiently large h .

Passing to the rational integral group ring of G , the absolutely irreducible representations of G are the absolutely irreducible representations of the simple components G_i of that semisimple algebra. Their centers are the fields of the corresponding characters, hence cyclotomic fields. By Theorem 3 it suffices, at the finitely many prime divisors of the fundamental ideal, to make the local degrees divisible by the corresponding indices. The field of n^h -th roots of unity does this for sufficiently large h .

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39. Hypercomplex Systems in Their Relations to Commutative Algebra and Number Theory

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1. The theory of hypercomplex systems, or algebras, has advanced strongly in recent years; but only very recently has the significance of this theory for commutative questions become clear. I want to report on this significance of the noncommutative for the commutative, and I shall do so by following two classical questions going back to Gauss: the principal genus theorem and the closely related norm theorem. Over time these questions have repeatedly changed their formulation. In Gauss they appear as the conclusion of the theory of quadratic forms; then they play an essential role in the characterization of relatively cyclic and abelian number fields by class field theory; finally they can be stated as theorems on automorphisms and on the splitting of algebras. This last formulation at the same time transfers the theorems to arbitrary relatively Galois number fields.

The principle of applying the noncommutative to the commutative is this: by means of the theory of algebras one seeks invariant and simple formulations for known facts about quadratic forms or cyclic fields, formulations depending only on structural properties of algebras. Once such invariant formulations are proved, the transfer to arbitrary Galois fields is obtained automatically.

2. First, an overview of the methods and further results. The main difficulty lies in finding the right formulation for general Galois fields; without the hypercomplex method there was no starting point for this. In the examples mentioned above, the underlying passage to the noncommutative is obtained by considering field and group simultaneously through the *crossed product* and its multiplication constants, the *factor systems*. Such crossed products were first considered by Dickson, whereas the theory of factor systems was developed by Speiser, Schur, and R. Brauer from a quite different starting point, namely the question of absolutely irreducible representations. Only by combining the two theories did one obtain a sufficiently simple and far-reaching structure.

At the same time this gives new and transparent proofs of known facts: Hasse's hypercomplex proof of the reciprocity law for cyclic fields by means of an invariant formulation of his norm residue symbol; and Chevalley's hypercomplex foundation of local class field theory on the same basis, for which new algebraic theorems on factor systems were needed. One must add the limitation that the crossed-product method by itself apparently does not yield the full theory of Galois number fields: new results of Artin, proceeding from Hasse's proof in the sense of this principle, give only numerical equalities in place of full isomorphism theorems.

Methods giving a full isomorphism, namely operator isomorphism, already exist in algebra. They continue ideas of A. Speiser: the Galois field is viewed as a *Galois module*, i.e. as a module over the base field which admits the substitutions of the Galois group as operators. There is an operator isomorphism between the field and the group ring in the sense that elements correspond one-to-one, linear forms over the base field correspond, and the substitutions of the Galois group in the field correspond to multiplication in the group ring. A theorem formulated by me was proved by M. Deuring, who based on it a proof of Galois theory. Further structural theorems of Deuring run parallel to formal facts of Artin L -series and give a structural approach to Artin conductors. These Artin L -series and conductors, formed with general group characters, together with Speiser's work, form the first connection between number theory and representation theory.

3. I now examine the norm theorem and the principal genus theorem in detail. First, the definition of the crossed product. Let K/k be a Galois field of degree n , with group G . The crossed product means a simultaneous embedding of K and G in an algebra A in such a way that the automorphisms of K become inner. Let u_S denote symbols corresponding to the n group elements. First set up A as a module of linear forms of rank n over K :

$$A = u_{S_1}K + \cdots + u_{S_n}K,$$

i.e. A consists of all linear forms $u_{S_1}a_1 + \cdots + u_{S_n}a_n$, $a_i \in K$. The inner automorphism condition gives

$$u_S^{-1}zu_S = z^S, \quad \text{or} \quad zu_S = u_S z^S \quad (z \in K),$$

and the multiplication is determined by

$$u_S u_T = u_{ST} a_{S,T}, \quad a_{S,T} \in K^*.$$

Associativity gives the factor-system condition

$$a_{S,T} a_{ST,R} = a_{S,TR} a_{T,R}^S.$$

The resulting A is called the crossed product of K with G with factor system $a_{S,T}$. One proves that A is a normal simple algebra over k , that K is a maximal commutative subfield, hence a splitting field, and conversely that for a given division algebra D there are always matrix rings D_r that can be obtained in this way as crossed products.

Replacing u_S by $v_S = u_S c_S$, $c_S \in K^*$, gives associated factor systems

$$a'_{S,T} = a_{S,T} \frac{c_S^T c_T}{c_{ST}}.$$

Associated factor systems are grouped into a class (a) ; similarly, all algebras similar to A are grouped into a class $\mathfrak{A}$. The classes $\mathfrak{A}$ and (a) correspond bijectively. The classes with fixed splitting field K form, under direct product, an abelian group isomorphic to the componentwise product of classes of factor systems. The identity element is the split algebra class, respectively the system of all transformation quantities

$$\frac{c_S^T c_T}{c_{ST}}.$$

4. For cyclic splitting fields this specializes to the norm concept. Let Z/k be cyclic, with S a generator of its group. One may let the powers of u correspond to the powers of S :

$$A = Z + uZ + \cdots + u^{n-1}Z,$$

$$zu = uz^S, \quad u^n = a, \quad a \in k^*,$$

and, if $v = uc$,

$$a' = aN(c).$$

Thus each factor system is represented by a single element $a \in k^*$, and one writes

$$A = (a, Z).$$

The identity class of factor systems is given by the norms from Z^* , so the group of algebra classes is isomorphic to

$$k^*/N(Z^*).$$

A cyclic algebra (a, Z) therefore splits if and only if a is the norm of an element of Z .

This connection between norm and splitting gives the formulation of the norm theorem as a theorem on split algebras: if an algebra splits at every place, then it splits absolutely. Here a place is defined, as in number theory, by replacing the base field k with its p -adic extension, and at infinite places by extension to the real numbers. For cyclic algebras this theorem says: if a is a local norm at every finite and infinite place, then a is the norm of a number of Z ; equivalently, without passing to p -adic fields, if a is a norm residue modulo every prime ideal and satisfies the known sign conditions, then a is a global norm. This is precisely the norm theorem of class field theory. The general theorem on split algebras follows from this cyclic special case by purely algebraic-arithmetic arguments. One first important consequence is Hasse's theorem: every simple normal algebra over an algebraic number field is cyclic.

5. A second consequence is the principal genus theorem. Its invariant formulation rests on the fact that the defining relations of the crossed product are purely multiplicative, and therefore still make sense when K^* is replaced by an abelian group $\mathfrak{G}$ whose automorphism group contains a subgroup isomorphic to G . The module of linear forms is then replaced by the extension of G by $\mathfrak{G}$ in the sense of group theory. If $\mathfrak{G}$ is taken to be the group of all ideals of K , the factor systems become systems of ideals. A division into classes in $\mathfrak{G}$ induces a division into classes for the factor systems.

The identity class of factor systems is defined to consist of those elements $a_{S,T}$ which generate split algebras at all finite and infinite ramified places of K . Let the corresponding extension of G be denoted by G^* . Then the invariant formulation of the principal genus theorem is:

If, under this class division, the substitutions

$$v_S = u_S(c_S)$$

produce an automorphism of G^* , that is, if the transformation quantities

$$\frac{(c_S)(c_T)^S}{(c_{ST})}$$

belong to the identity ideal class of factor systems, then there is an ideal class $(\mathfrak{b})$ such that

$$(c_S) = \frac{(\mathfrak{b})}{(\mathfrak{b})^S}$$

for all $S \in G$. The hypothesis expresses the automorphism property; the fact that this automorphism is inner is expressed by

$$v_S = (\mathfrak{b})^{-1}u_S(\mathfrak{b}) = u_S(\mathfrak{b})^{1-S} = u_S(c_S).$$

In the cyclic special case one obtains: if $N([c])$ lies in the identity ideal class of factor systems, then (c) is a symbolic $(1 - S)$ -st power,

$$(c) = (\mathfrak{b})^{1-S}.$$

6. The theorem has been stated for full ideal classes; its content is unchanged if, as usual, one restricts to ideals prime to the ramified places. With this restriction, the principal genus defined here for quadratic fields becomes Gauss's principal genus: ideal classes correspond to quadratic forms, and the norms of classes correspond to the numbers represented by the forms. The identity class means that these represented numbers are quadratic residues at the ramified places; the associated forms therefore have the total character of the principal form and constitute Gauss's principal genus. It is known that the symbolic $(1 - S)$ -st power becomes duplication.

For cyclic fields the statement becomes: the principal genus consists of all ideal classes whose norms are norm residues at the finite and infinite ramified places. This is equivalent to being a norm residue modulo the conductor, and the usual theorem is obtained as a specialization.

Even for arbitrary Galois fields one can introduce a conductor composed only of the ramified places, in such a way that after normalizing the factor systems, at each such place the ray contains only elements of the identity class. This raises the question of the connection with Artin conductors, and hence with the theory of Galois modules, the second hypercomplex method. The future must show how far these two methods reach.